GENERATIVE AI

GANs: Generative Adversarial Networks

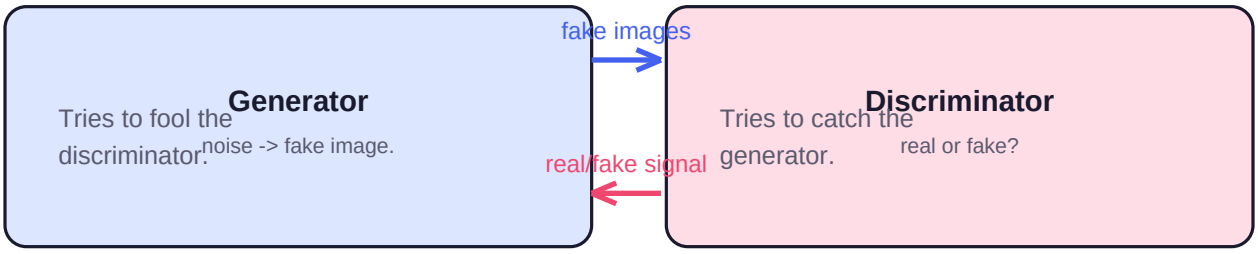
Two networks, one game:
generator vs discriminator, training dynamics, failure modes.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

The Adversarial Game

A forger and a detective, locked in competition.

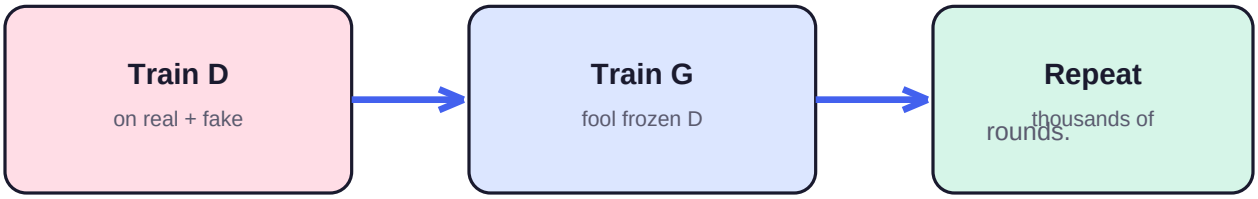


Minimax game: the generator minimizes what the discriminator maximizes. Equilibrium = fakes indistinguishable from real.

STEP 2/5

Training Dynamics

Alternate updates; balance is everything.



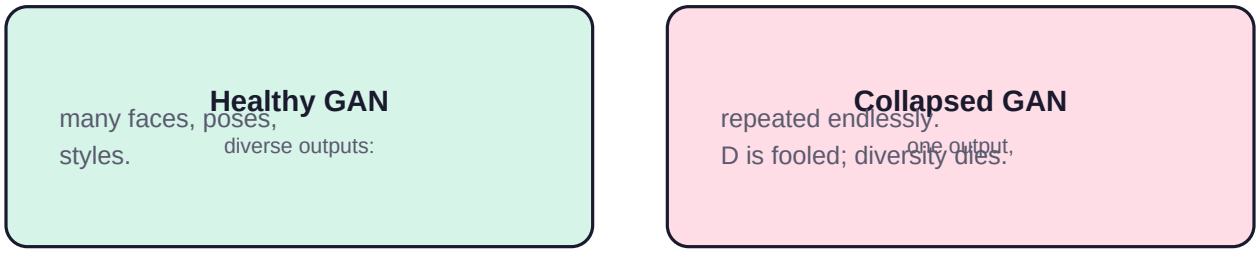
Freeze one network while updating the other. If D gets too strong, G gets no useful gradient.

GAN training is famously unstable: learning rates, update ratios, and architectures must be tuned together.

STEP 3/5

Mode Collapse

The generator finds one trick and repeats it.



Fixes: minibatch discrimination, unrolled GANs, Wasserstein loss, or simply better architectures (StyleGAN). Diversity metrics catch it early.

STEP 4/5

Evaluating GANs

No loss curve tells you the images are good.

- FID (Frechet Inception Distance): compares feature distributions of real vs generated images — lower is better.
- Inception Score: rewards sharp, diverse classifications — higher is better, but gameable.
- Precision/Recall for GANs: precision = quality of fakes, recall = coverage of real diversity.
- Human eval still matters: show raters mixed real/fake grids and measure fool rate.
- Track FID through training; stop or roll back when it degrades — GANs can collapse late.

STEP 5/5

Modern Variants

Where GAN ideas live on.

StyleGAN

Style-based generator: control pose, texture, identity separately. Photoreal faces.

CycleGAN

Unpaired image-to-image translation: horses to zebras, summer to winter.

Conditional GANs

Feed the label in: generate 'a cat', not just 'an image'.

Diffusion era

Diffusion models largely replaced GANs for quality — GANs still win on single-step speed.

STEP ✓/✓

Key Takeaways

GANs — Remember

- Generator forges, discriminator detects: a minimax game toward indistinguishability.
- Train by alternating updates; keep the two networks balanced.
- Mode collapse = one repeated output; watch diversity metrics, not just loss.
- Evaluate with FID + precision/recall + human fool rates.
- StyleGAN/CycleGAN/conditional variants; diffusion now leads on quality, GANs on speed.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev